

0. Real numbers and functions

Absolute Value

Properties

1. $|xy| = |x||y| \quad \forall x, y \in \mathfrak{R}$
2. $\sqrt{x^2} = |x| \Leftrightarrow |x|^2 = x^2 \quad \forall x \in \mathfrak{R}$
3. $|x + y| \leq |x| + |y| \quad \forall x \in \mathfrak{R}$ (Triangle Inequality)

Example

Solve $|x + 1| \leq |2x - 1|$. (Algebraically)

$$|x + 1| \leq |2x - 1| \quad \text{Given}$$

$$|x + 1|^2 \leq |2x - 1|^2 \quad \text{Both expressions are positive}$$

$$x^2 + 2x + 1 \leq 4x^2 - 4x + 1$$

$$3x^2 - 6x \geq 0$$

$$3x(x - 2) \geq 0$$

$$x \leq 0 \vee x \geq 2$$

Functions

Let a function be $f : A \rightarrow B$. f is bijective if it is both injective and surjective.

Injective

For any $x, y \in A$, $f(x) = f(y) \Rightarrow x = y$. (One-one)

Surjective

For any $z \in B$, $\exists x \in A$ where $f(x) = z$. (Range = Co-Domain)

Note that range $\neq$ co-domain. Co-domain is potential outputs of f (in the definition); Range is the actual outputs of f .

Inverse Functions

If f is bijective, $\exists f^{-1} : B \rightarrow A$.

Why?

1. If f is not injective, the inverse function would have more than 1 output for 1 input which violates function definition.
2. If f is not surjective, the inverse function would have certain values in B that have no mapping (undefined).

Maximal domain/range

The largest domain/range that the function can have in the context of real numbers $\mathfrak{R}$.

Eg. $f(x) = \frac{1}{x-1}$; maximal domain $D = \mathfrak{R} \setminus \{1\}$; maximal range $R = \mathfrak{R} \setminus \{0\}$

1. Limits

Properties

1. Let c be an interior point (not an end point). If $\lim_{x \rightarrow c^-} f(x) = \lim_{x \rightarrow c^+} f(x) = L \Rightarrow \exists \lim_{x \rightarrow c} f(x) = L$. It is said that $f(x)$ is continuous at c .
2. Continuity:
 - Left end point: $\exists \lim_{x \rightarrow c^+} f(x) = f(c)$
 - Right end point: $\exists \lim_{x \rightarrow c^-} f(x) = f(c)$
 - Interior point: $\exists \lim_{x \rightarrow c} f(x) = f(c)$
 - On an interval I : $f(x)$ is continuous $\forall c \in I$.
3. Results:
 - $\lim_{x \rightarrow c} [f(x) \pm g(x)] = \lim_{x \rightarrow c} f(x) \pm \lim_{x \rightarrow c} g(x)$
 - $\lim_{x \rightarrow c} kf(x) = k \lim_{x \rightarrow c} f(x)$
 - $\lim_{x \rightarrow c} [f(x)g(x)] = \lim_{x \rightarrow c} f(x) \cdot \lim_{x \rightarrow c} g(x)$
 - $\lim_{x \rightarrow c} [f(x)]^2 = [\lim_{x \rightarrow c} f(x)]^2$
 - $\lim_{x \rightarrow c} \left[\frac{f(x)}{g(x)} \right] = \frac{\lim_{x \rightarrow c} f(x)}{\lim_{x \rightarrow c} g(x)}$
 - If g is continuous at point b and $\lim_{x \rightarrow c} f(x) = b$, then $\lim_{x \rightarrow c} g \odot f(x) = g(b)$.
4. Limits of form $\frac{\sin x}{x}$ and $\frac{\tan x}{x}$

If $\lim_{x \rightarrow c} g(x) = 0$,

$$\lim_{x \rightarrow c} \left(\frac{\sin g(x)}{g(x)} \right) = 1$$

$$\lim_{x \rightarrow c} \left(\frac{\tan g(x)}{g(x)} \right) = 1$$

Squeeze theorem

Suppose I is an interval that contains point c and $g(x) \leq f(x) \leq h(x)$ for x near c .

$$\lim_{x \rightarrow c} g(x) = \lim_{x \rightarrow c} h(x) = L \Rightarrow \lim_{x \rightarrow c} f(x) = L$$

Example

Evaluate $\lim_{x \rightarrow \infty} \frac{\sin(2 \ln x)}{\ln x}$.

Since $-1 \leq \sin(2 \ln x) \leq 1$,

$$\begin{aligned} -\frac{1}{\ln x} &\leq \frac{\sin(2 \ln x)}{\ln x} \leq \frac{1}{\ln x} \\ \lim_{x \rightarrow \infty} -\frac{1}{\ln x} &\leq \lim_{x \rightarrow \infty} \frac{\sin(2 \ln x)}{\ln x} \leq \lim_{x \rightarrow \infty} \frac{1}{\ln x} \\ 0 &\leq \lim_{x \rightarrow \infty} \frac{\sin(2 \ln x)}{\ln x} \leq 0 \\ \therefore \lim_{x \rightarrow \infty} \frac{\sin(2 \ln x)}{\ln x} &= 0 \end{aligned}$$

Intermediate value theorem

Suppose f is continuous on $[a, b]$ and $f(a) \leq k \leq f(b)$. This implies that $\exists c \in [a, b]$, $f(c) = k$.

Example

Show $x^3 e^x = 10$ has a solution between 1 and 1.5.

Let $f(x) = x^3 e^x$.

$$f(1) = 1^3 \cdot e^1 = e$$

$$f(1.5) = 1.5^3 \cdot e^{1.5} = 15.1$$

$$\Leftrightarrow f(1) \leq 10 \leq f(1.5)$$

$$\therefore \exists f(c) = 10, c \in [1, 1.5]$$

2. Derivatives

Definition

$$\begin{aligned} f'(x_0) &= \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h} \\ &= \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} \text{ by letting } x = x_0 + h \end{aligned}$$

Gradient of chord of $(x_0, f(x_0))$ and $(x_0 + h, f(x_0 + h))$ approaches the tangent of f .

If f is differentiable on every point on I , it is said to be differentiable on I .

Inverse functions

Let f be bijective and differentiable on I . Then,

$$(f^{-1})'(a) = \frac{1}{f'(f^{-1}(a))}$$

Proof: differentiate $f^{-1}(f(x)) = x$ with chain rule.

3. Applications of differentiation

Straight line equation

$$y - y_1 = m(x - x_1)$$

Concavity

1. If $f''(c) > 0$, f is concave up at c .
2. If $f''(c) < 0$, f is concave down at c .

Extreme value theorem

If f is continuous on $[a, b]$, then $\exists$ a local maximum and minimum in the range $[a, b]$.

Note that this will not hold if f is not continuous on $[a, b]$ or interval is not of the form $[a, b]$ (e.g. $(a, b]$)

Critical point

If f has a local minimum/maximum at $x = c$, c is a **critical point** of f .

The following conditions must be fulfilled:

1. c is **not** an endpoint
2. $f'(c) = 0$ or $\nexists f'(c)$

Local and global maxima/minima

- Global: if a is the global minima/maxima, there is no point b at which $f(b) > f(a) \forall f \in D_f$.
- Local: if a is the local minima/maxima, there may be a point b (perhaps outside of the target interval) where $f(b) > f(a)$

L'Hopital's rule

If $\lim_{x \rightarrow c} g(x) = \lim_{x \rightarrow c} f(x) = 0 \vee \pm\infty$,

$$\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \lim_{x \rightarrow c} \frac{f'(x)}{g'(x)}$$

Rolle's theorem

Let f be differentiable on $[a, b]$, where $f(a) = f(b)$. Then there $\exists c \in [a, b]$ where $f'(c) = 0$.

Mean Value Theorem

Let f be differentiable on $[a, b]$. There $\exists c \in [a, b]$ where $f'(c) = \frac{f(b) - f(a)}{b - a}$.

This is because the chord between a and b is parallel to the tangent at c .

4. Integrals

Techniques: IBP, u-substitution, trig-substitution, partial fractions

Trig sub

$$\int \sqrt{a^2 - (x+b)^2} dx \text{ (sub with } (x+b) = a \sin \theta)$$

$$\int \sqrt{a^2 + (x+b)^2} dx \text{ (sub with } (x+b) = a \tan \theta)$$

$$\int \sqrt{(x+b)^2 - a^2} dx \text{ (sub with } (x+b) = a \sec \theta)$$

Fundamental theorem of calculus

FTC 1

$$\int_b^a f(x) dx = F(a) - F(b)$$

FTC 2

$$\frac{d}{dx} \int_a^x f(t) dt = f(x); \text{ assuming continuity and differentiability for } a \leq x \leq b$$

Riemann sums

Define single integral with:

$$\int_b^a f(x) dx = \lim_{n \rightarrow \infty} \left\{ \sum_{k=1}^n \frac{b-a}{n} f\left(a + k\left(\frac{b-a}{n}\right)\right) \right\}$$

Where $\frac{b-a}{n}$ is the small dx of the function and $f\left(a + k\left(\frac{b-a}{n}\right)\right)$ is the height of the small increment.

5. Applications of Integration

Area between 2 curves

$$A = \int_b^a |f(x) - g(x)| \, dx$$

Volume formed by rotation

Shell method

$$V = \pi \int_b^a (f(x))^2 dx - \pi \int_b^a (g(x))^2 dx$$

Disk method

For volume rotated about y axis, consider cylinder with circumference $2\pi x$, height $f(x)$. Volume is the product of circumference and the riemann sum of $f(x)$.

$$V = 2\pi \int_b^a x |f(x) - g(x)| \, dx$$

Arc Length of curve

Consider small slanted line segments forming $f(x)$. By Pythagoras' theorem, $ds = \sqrt{(dx)^2 + (dy)^2} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx = \sqrt{1 + f'(x)^2} dx$. Then,

$$S = \int_b^a \sqrt{1 + (f'(x))^2} dx$$

6. Sequences and series

Limits of sequence

If a_n and b_n are convergent (ie. limit of sequence as $n \rightarrow \infty$ exists as a real number), we have the same results as 1.3.

Series tests

nth term for divergence

If $\lim_{n \rightarrow \infty} a_n \neq 0 \vee$ undefined, then series $\sum_{n=1}^{\infty} a_n$ diverges. Inconclusive if $a_n \rightarrow 0$.

integral test

Suppose $a_n = f(n)$; $f'(n) < 0 \forall n > 1$ (decreasing function). If $\int_1^{\infty} f(n)$ converges, so does the series $\sum_1^{\infty} a_n$, and vice-versa. (Reasoning: think of right bounded riemann sum to approximate integral)

The p-series $\sum_1^{\infty} \frac{1}{n^p}$ converges only if $p > 1$.

comparison test

Suppose $0 \leq a_n \leq b_n$ (b_n is the upper bound). Then,

1. $\sum_1^{\infty} b_n$ is convergent $\Rightarrow \sum_1^{\infty} a_n$ converges
2. $\sum_1^{\infty} b_n$ is divergent $\Rightarrow \sum_1^{\infty} a_n$ diverges

We often compare to a p-series (eg. $b_n = \frac{1}{n^k}, k > 1$) or geometric series.

ratio test

Suppose $\sum_1^{\infty} a_n$, where $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = L$. If,

1. $0 \leq L < 1$, series converge
2. $L > 1$, series diverge
3. $L = 1$, test inconclusive

ie. compare the ratio of the (n+1)th term to the nth term.

Note: remember for convergence/finding the radius of convergence, create an inequality such that the final term is less than 1.

root test

Suppose $\sum_1^{\infty} a_n$, where $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = L$. If,

1. $0 \leq L < 1$, series converge
2. $L > 1$, series diverge
3. $L = 1$, test inconclusive

Test functions of the form $(a_n)^{f(n)}$, easier to reduce.

Alternating series

If b_n is decreasing and $\lim_{n \rightarrow \infty} b_n = 0$, then alternating series $\sum_1^{\infty} (-1)^n b_n$ converges.

Absolute convergence

If $\sum_{n=1}^{\infty} |a_n|$ is convergent, $\sum_{n=1}^{\infty} a_n$ is also convergent.

Power series

For a given series $\sum_{n=1}^{\infty} c_n(x-a)^n = c_0 + c_1(x-a) + c_2(x-a)^2 + c_3(x-a)^3 + \dots$, **exactly 1** of the following holds:

1. Converge at $x = a$

2. Converge for all x
3. There exists a positive R where $|x - a| < R$ converges (radius of convergence)

Note: radius of convergence may take a bounded or unbounded form, eg $(-1, 1]$ or $[-1, 1]$

Note: centre of convergence is a .

Note: Use ratio/root test to find R .

Finding R via ratio test or root test

If $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = L(x - a)$ (ratio test) or $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = L(x - a)$ (root test), $R = \frac{1}{L}$.

Note: this method only proves for the unbounded form. check the limits as x approaches the endpoints to prove boundedness.

Power series representation

If $R > 0$, $\sum_{n=1}^{\infty} c_n(x - a)^n$ is differentiable and integrable on interval $|x - a| < R$.

Note: $\frac{1}{1-x} = \sum_{n=1}^{\infty} x^n$.

eg. Find power series representation of $\tan^{-1} x$.

$$\begin{aligned} \frac{1}{1+x^2} &= \sum_{n=1}^{\infty} (-x^2)^n = \sum_{n=1}^{\infty} (-1)^n x^{2n} \\ \int \frac{1}{1+x^2} dx &= \int \sum_{n=1}^{\infty} (-1)^n x^{2n} dx \\ \tan^{-1} x + C_1 &= \sum_{n=1}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1} + C_2 \end{aligned}$$

Note: substitute $x=0$ to find C .

Taylor, Maclaurin Series

Taylor series: $\sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x - a)^n$

7. Vectors!

Dot product

1. $a \cdot b = b \cdot a$ (commutativity)
2. $a \cdot (b + c) = a \cdot b + a \cdot c$ (distributive law)
3. $(da) \cdot b = d(a \cdot b) = a \cdot (db)$
4. $0 \cdot a = 0$
5. $a \cdot a = |a|^2$

Cross product

1. $a \times b = -b \times a$
2. $(da) \times b = d(a \times b) = a \times (db)$
3. $a \times (b + c) = a \times b + a \times c$
4. $(a + b) \times c = a \times c + b \times c$

Projection

Component of b along a : $\text{comp}_a b = |b| \cos \theta = \frac{a \cdot b}{|a|}$

Projection of b onto a : $\text{proj}_a b = \text{comp}_a b \times \frac{a}{|a|}$ (multiply by unit vector)

Line, Plane

Parametric straight line: $x = x_0 + at, y = y_0 + bt, z = z_0 + ct$

Parametric plane: $ax + by + cz = d$

x, y, z are any point on the line. a, b, c are any vector parallel to the line.

8. Multivariable functions

Definitions

- Vector functions of 1 variable: $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$
- Functions of 2 variables: $z = f(x, y)$
 - use a 3d graph (will look like a 3d volume)/contour plot (drawn on 2d using many values of z).
 - domain is 2d (x, y)
 - range is 1d (z)
- Cylinder: A surface is a cylinder if there is a plane P such that all the planes parallel to P intersect the surface in the same curve (when viewed in 2-dimension)
 - ie. any equation in x, y, z and missing 1 variable is a cylinder
- Functions of 3 variables: $w = f(x, y, z)$
 - Level surface: $k = f(x, y, z)$ for some constant k . Draw $f(x, y, z)$ as k changes, will appear as shifting 3d surfaces.

Partial derivatives

Treat x or y as a constant and differentiate.

$$\begin{aligned} f_x(x, y) &= \lim_{h \rightarrow 0} \frac{f(x+h, y) - f(x, y)}{h} \\ f_y(x, y) &= \lim_{h \rightarrow 0} \frac{f(x, y+h) - f(x, y)}{h} \end{aligned}$$

Clairaut's theorem

f is defined on a disk D (2d xy plane as domain) that contains (a, b) .

If $f_{xy}(x, y)$ and $f_{yx}(x, y)$ are continuous on D , $f_{xy}(x, y) = f_{yx}(x, y)$.

Tangent planes

Consider a point on a surface $P(a, b, c)$. The tangent plane at P contains lines passing through P with the gradient given by the partial derivatives of x, y at P .

In a y -plane (y is constant), a vector parallel to plane is $\begin{pmatrix} 1 \\ 0 \\ f_x(a, b) \end{pmatrix}$

In an x -plane (x is constant), a vector parallel to plane is $\begin{pmatrix} 0 \\ 1 \\ f_y(a, b) \end{pmatrix}$

Note: change in x of 1 unit corresponds to change in z by $f_x(a, b)$ (y does not change as it is in the y -plane). Same for y .

Hence vector normal to the plane $\begin{pmatrix} 1 \\ 0 \\ f_x(a, b) \end{pmatrix} \times \begin{pmatrix} 0 \\ 1 \\ f_y(a, b) \end{pmatrix} = \begin{pmatrix} f_x(a, b) \\ f_y(a, b) \\ -1 \end{pmatrix}$.

Then, substitute P to get the equation of the plane. $\mathbf{n} \cdot \begin{pmatrix} x-a \\ y-b \\ z-c \end{pmatrix} = 0$

$$z = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$$

Note:

- The vector from $P(a, b, c)$ to any point (x, y, z) on the plane is: $\begin{pmatrix} x-a \\ y-b \\ z-c \end{pmatrix}$
- This vector lies in the plane
- Therefore, it must be perpendicular to the normal vector $\mathbf{n}$ (ie. dot product returns 0)

Differentiation of multivariable functions

Chain rule

If $z = f(x, y)$ and $x = f(t)$ and $y = g(t)$ (single variable t),

$$\frac{dz}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt}$$

In general,

$$\frac{\partial u}{\partial t_i} = \sum_{j=1}^n \frac{\partial u}{\partial x_j} \frac{\partial x_j}{\partial t_i} = \frac{\partial u}{\partial x_1} \frac{\partial x_1}{\partial t_i} + \frac{\partial u}{\partial x_2} \frac{\partial x_2}{\partial t_i} + \dots + \frac{\partial u}{\partial x_n} \frac{\partial x_n}{\partial t_i}$$

Implicit differentiation

If $F(x, y, z) = 0$ defines z implicitly as a differentiable function of x and y , then:

$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z}, \quad \frac{\partial z}{\partial y} = -\frac{F_y}{F_z}$$

provided $F_z \neq 0$.

Reasoning: Differentiate $F(x, y, z) = 0$ with respect to x using the chain rule:

$$\frac{\partial F}{\partial x} \frac{\partial x}{\partial x} + \frac{\partial F}{\partial y} \frac{\partial y}{\partial x} + \frac{\partial F}{\partial z} \frac{\partial z}{\partial x} = 0$$

Since x and y are independent variables, $\frac{\partial x}{\partial x} = 1$ and $\frac{\partial y}{\partial x} = 0$. This simplifies to:

$$F_x + F_z \frac{\partial z}{\partial x} = 0$$

Solving for $\frac{\partial z}{\partial x}$ (assuming $F_z \neq 0$):

$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z}$$

The same process applied with respect to y yields $\frac{\partial z}{\partial y} = -\frac{F_y}{F_z}$.

Increments and Differentials

Increment

Let $z = f(x, y)$. Suppose Δx and Δy are increments in the independent variables x and y respectively.

Then the **increment** in z is defined by:

$$\Delta z = f(x + \Delta x, y + \Delta y) - f(x, y)$$

Differential

Let $z = f(x, y)$. Suppose Δx and Δy are increments in the independent variables x and y respectively.

Then the **differentials** of the independent variables x and y are:

$$dx = \Delta x, \quad dy = \Delta y$$

The **differential** (or **total differential**) of the dependent variable z is:

$$dz = f_x(x, y)dx + f_y(x, y)dy$$

Key Distinction:

- The increment Δz is the change in z as (x, y) changes from (a, b) to $(a + \Delta x, b + \Delta h)$.
- The differential dz is the change in the **tangent plane** as (x, y) changes from (a, b) to $(a + \Delta x, b + \Delta h)$.

Differential approximation

Suppose f is differentiable at (a, b) . Let Δx and Δy be small increments in x and y respectively from (a, b) . Then:

$$\Delta z \approx dz = f_x(a, b)\Delta x + f_y(a, b)\Delta y = f_x(a, b)dx + f_y(a, b)dy$$

provided Δx and Δy are small and $f(x, y)$ is differentiable.

Directional Derivatives and Gradient

Directional Derivative

The **directional derivative** of $f(x, y)$ at (x_0, y_0) in the direction of unit vector $\mathbf{u} = \langle a, b \rangle$ is:

$$D_{\mathbf{u}}f(x_0, y_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + ha, y_0 + hb) - f(x_0, y_0)}{h}$$

provided this limit exists.

Computing Directional Derivative

If $f(x, y)$ is a differentiable function, then f has a directional derivative in the direction of any unit vector $\mathbf{u} = \langle a, b \rangle$ and:

$$D_{\mathbf{u}}f(x, y) = f_x(x, y)a + f_y(x, y)b = \nabla f(x, y) \cdot \mathbf{u}$$

(shortcut: use ∇f to compute)

Gradient

The **gradient** of $f(x, y)$ is the vector-valued function:

$$\nabla f(x, y) = \langle f_x, f_y \rangle = f_x \mathbf{i} + f_y \mathbf{j} = \frac{\partial f}{\partial x} \mathbf{i} + \frac{\partial f}{\partial y} \mathbf{j}$$

provided both partial derivatives exist.

3-D Directional Derivative

The **directional derivative** of $f(x, y, z)$ at (x_0, y_0, z_0) in the direction of unit vector $\mathbf{u} = \langle a, b, c \rangle$ is:

$$D_{\mathbf{u}}f(x_0, y_0, z_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + ha, y_0 + hb, z_0 + hc) - f(x_0, y_0, z_0)}{h}$$

provided this limit exists.

Note: $\mathbf{u}$ is a unit vector to standardise change to be 1 unit. If requirement is n units, multiply final result by n .

Computing 3-D Directional Derivative

$$D_{\mathbf{u}}f(x_0, y_0, z_0) = \nabla f(x_0, y_0, z_0) \cdot \mathbf{u}$$

where

$$\nabla f = \langle f_x, f_y, f_z \rangle = \frac{\partial f}{\partial x} \mathbf{i} + \frac{\partial f}{\partial y} \mathbf{j} + \frac{\partial f}{\partial z} \mathbf{k}$$

is the **gradient vector**.

Increase/decrease

$$D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u} = |\nabla f| \cos \theta$$

since $\mathbf{u}$ is a unit vector.

- **Maximum rate of change:** When $\cos \theta = 1$
- **Minimum rate of change:** When $\cos \theta = -1$

Local extrema

If f has a local maximum or minimum (a, b) , then

$$f_x(a, b) = f_y(a, b) = 0$$

Saddle points

Also a critical point, but not a local minimum/maximum.

Note: Extrema are “decisive” – the function goes up or down in all directions; while saddle points are “indecisive” – they’re minima in some directions and maxima in others.

Second derivative test

Suppose $f(x, y)$ has continuous second-order partial derivatives on some open disk centered at (a, b) . Suppose $f_x(a, b) = f_y(a, b) = 0$ (that is (a, b) is a critical point). Define the *discriminant* D for the point (a, b) by

$$D = D(a, b) = f_{xx}(a, b)f_{yy}(a, b) - [f_{xy}(a, b)]^2$$

1. If $D > 0$ and $f_{xx}(a, b) > 0$, then $f(a, b)$ is a local minimum.
2. If $D > 0$ and $f_{xx}(a, b) < 0$, then $f(a, b)$ is a local maximum.
3. If $D < 0$, then (a, b) is a saddle point of f .
4. If $D = 0$, then no conclusion can be drawn.

Double integrals

The **double integral** of f over the rectangle R is

$$\int \int_R f(x, y) \, dA = \lim_{m, n \rightarrow \infty} \sum_{i=1}^m \sum_{j=1}^n f(x_{ij}^*, y_{ij}^*) \Delta A$$

provided the limit exists and is the same for any choice of the sample points (x_{ij}^*, y_{ij}^*) in R_{ij} , for $1 \leq i \leq m, 1 \leq j \leq n$.

When this happens, we say that f is **integrable** over R .

Fubini's Theorem

If f is continuous on the rectangle $R = \{(x, y) \mid a \leq x \leq b, c \leq y \leq d\}$, then

$$\int \int_R f(x, y) \, dA = \int_a^b \int_c^d f(x, y) \, dy \, dx = \int_c^d \int_a^b f(x, y) \, dx \, dy$$

In other words, the double integral can be computed as an iterated integral in either order.

Splitting (special case)

If $f(x, y) = g(x)h(y)$,

$$\int \int_R g(x)h(y) \, dA = \left(\int_a^b g(x) \, dx \right) \left(\int_c^d h(y) \, dy \right)$$

where $R = [a, b] \times [c, d]$

Double integral over general region

If f is continuous on a Type I domain D such that

$$D = \{(x, y) \mid a \leq x \leq b, g_1(x) \leq y \leq g_2(x)\}$$

then

$$\int \int_D f(x, y) \, dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) \, dy \, dx.$$

If f is continuous on a Type II domain D such that

$$D = \{(x, y) \mid c \leq y \leq d, h_1(y) \leq x \leq h_2(y)\}$$

then

$$\int \int_D f(x, y) \, dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x, y) \, dx \, dy.$$

Polar coordinates

If f is continuous on a polar rectangle R given by

$$R = \{(r, \theta) \mid 0 \leq a \leq r \leq b, \alpha \leq \theta \leq \beta\}$$

where $0 \leq \beta - \alpha \leq 2\pi$, then

$$\int \int_R f(x, y) \, dA = \int_\alpha^\beta \int_a^b f(r \cos \theta, r \sin \theta) r \, dr \, d\theta.$$

Note: remember the $rdrd\theta$

Determining Bounds for Polar Coordinates

To convert a double integral to polar coordinates:

1. **For r bounds:** Draw a ray from the origin at angle θ . The ray enters the region at $r = r_1(\theta)$ and exits at $r = r_2(\theta)$. These become your limits for r .
2. **For θ bounds:** Determine the range of angles needed to sweep through the entire region. This gives your limits for θ .

The Jacobian and Change of Variables

For a transformation $x = x(u, v)$ and $y = y(u, v)$, the **Jacobian** is:

$$J = \frac{\partial(x, y)}{\partial(u, v)} = \det \begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{pmatrix} = \frac{\partial x}{\partial u} \frac{\partial y}{\partial v} - \frac{\partial x}{\partial v} \frac{\partial y}{\partial u}$$

The change of variables formula is:

$$\int \int_R f(x, y) \, dx \, dy = \int \int_S f(x(u, v), y(u, v)) |J| \, du \, dv$$

The Jacobian measures how the transformation distorts area. A small region with area $du \cdot dv$ in the uv -plane maps to area $|J| \cdot du \cdot dv$ in the xy -plane, which is why we multiply by $|J|$ when changing variables.

Surface Area

Let $\overrightarrow{PQ}$ be the vector on the tangent plane at P with x -component dx , and $\overrightarrow{PR}$ the vector with y -component dy . Thus, $\overrightarrow{PQ} = \langle dx, 0, f_x(x, y)dx \rangle$ and $\overrightarrow{PR} = \langle 0, dy, f_y(x, y)dy \rangle$. The area of the parallelogram spanned by $\overrightarrow{PQ}$ and $\overrightarrow{PR}$ is the magnitude of the cross product $\overrightarrow{PQ} \times \overrightarrow{PR}$.

$$\overrightarrow{PQ} \times \overrightarrow{PR} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ dx & 0 & f_x dx \\ 0 & dy & f_y dy \end{vmatrix} = (-f_x, -f_y, 1) dx dy.$$

Therefore, $dS = |(-f_x, -f_y, 1) dx dy| = \sqrt{f_x^2 + f_y^2 + 1} dA$.

$$\text{Surface Area} = \int \int_D dS = \int \int_D \sqrt{f_x^2 + f_y^2 + 1} dA$$

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Differential Equations

Separable DE

Such an equation is of the form $y' = f(x)g(y)$.

Methods to separate:

1. $y' = f\left(\frac{y}{x}\right)$: use $u = \frac{y}{x} \Rightarrow y' = (ux)' = u + u'x$
2. $y' = f(ax + by)$: use $u = ax + by$.

Linear first order ODE

Given form $\frac{dy}{dx} + P(x)y = Q(x)$, define $I(x) = e^{\int P(x)dx}$. Then,

$$y \cdot I(x) = \int Q(x) \cdot I(x) dx$$

Bernoulli equation

Given form $y' + P(x)y = Q(x)y^n$, substitute $u = y^{1-n}$.

Then get $u' + (1-n)P(x)u = (1-n)Q(x)$, which is first order ODE.